

RESEARCH INTERESTS

- Computer Vision and GenAI
 - Synthetic data generation and segmentation
 - Vision language modeling
 - Predictive modeling using causal analysis
 - Deployment on low-memory devices
- IoT and Simulation
 - Lightweight cloud-edge frameworks
 - Operational digital twin, Human digital twin
 - Augmented / Virtual / Extended Reality
 - Human - machine interaction

EDUCATION

Ph.D., University of Cincinnati, OH, USA. GPA 4.0 2018 – Present
Dissertation: Machine learning and industrial IoT-assisted computer vision frameworks for smart and safe manufacturing (Advisor: Prof. Sam Anand)

M.S., Mechanical Engineering, Michigan Technological University, MI, USA. GPA 3.95 2015 – 2017

B.S., Mechanical Engineering, University of Pune, India. GPA 3.90 2010 – 2014

WORK EXPERIENCE

Graduate Research Assistant, University of Cincinnati 2020 – Present

LLM-based human motion generation

Industry collaborators – UC Industry 4.0/5.0 Institute – [P&G](#), [Siemens](#), [Worthington Steel et al.](#)

Conditioned on natural language 2024

- Devised machine-in-the-loop framework to collect motion training data from human experts
- Proposed a novel loss function to train a GPT model that generates ergonomic-friendly motions

Conditioned on prior sequence of motions 2024

- Developed autoregressive model to generate projected trajectory for providing safety alerts
- Developed framework to convert 3D pose estimates to SMPL using keypoint conversion

IIOT for legacy and smart factory machines with LLM feedback 2024

Industry collaborators – UC Industry 4.0/5.0 Institute - [Siemens](#), [Worthington Steel et al.](#)

- Collaborated on a RAG-based multi-modal multi-agent LLM for real-time control of factory machines
- Proposed a pipeline to extract and visualize metadata through a digital twin

Deep learning-based component recognition enabled in augmented reality for worker training 2023

Industry collaborators – UC Industry 4.0/5.0 Institute - [P&G](#), [Siemens](#), [Kinetic Vision et al.](#)

- Generated CAD synthetic image segmentation data using physics-based rendering methods
- Fine-tuned U-Net architecture and deployed it on an AR app to highlight objects in real-time

Real-time sealant deposition monitoring for automated quality control using computer vision 2022

Industry collaborators – [Siemens Technology and Boeing](#)

- Performed object localization and segmentation on images captured by a robotic arm
- Trained a Bayesian network to predict the real-time sealant parameters for space-grade solar panel

Integration of Siemens IIoT tools for smart factory framework and analytics 2021

Industry collaborator – [Siemens Digital Industries](#)

- Created mixed-reality apps for real-time factory data monitoring by ensuring minimal latency
- Presented a white paper in Siemens Realize Live 2022

CAD-based computational tools for metal powder additive manufacturing

Industry collaborator – [Raytheon Technologies](#) 2020

- Developed computational geometry algorithms for feature recognition in 3D geometries
- Designed frontend application GUI within Siemens NX

Engineering Intern – Research and Development

2015

Industry – Tata Motors, India

- Designed rapid prototype components in CAD for 3D printing
- Designed sheet metal components in CAD for bending operations

RELEVANT RESEARCH PROJECTS

Deep learning segmentation methods for defect detection

2024

- Published a survey of CV methods reported for 3D printing using YOLO, SSD, Faster R-CNN, U-Net, Mask R-CNN, GAN, and Vision Transformers and implementing them for real-time monitoring

Immersive visualization methods in manufacturing

2023

- Developed a digital twin through an IoT framework with API, cloud-MQTT interface for real-time human design interaction, and control machine parameters from virtual reality with authentication

Prediction of selective laser melting (3D printing) part quality using hybrid Bayesian network

2019

- Devised a Bayesian network containing discrete and continuous nodes with a novel n-dimensional convex hull for fast prediction of 3D printed part quality

PROGRAMMING AND SOFTWARE SKILLS

Programming languages: Python, C#, C++, MATLAB, JavaScript, HTML, SQLite

Developer tools / IDE: PyCharm, Visual Studio Code, Jupyter Notebook

Commercial software: NX, Siemens Jack/Process Simulate, Ansys Additive, Autodesk Netfabb, Cortex

Open-source libraries: OpenCV, PyTorch, Tensorflow, Blender, ONNX, GitHub, HuggingFace

PUBLICATIONS

Deep learning-based recognition of manufacturing components using augmented reality for worker training of assembly tasks

2024

Sourabh Deshpande, Manish Aryal, Sam Anand

ASME Manufacturing Science and Engineering Conference, <https://doi.org/10.1115/MSEC2024-125279>

ImVR: Enabling immersive design exploration and process integration for additive manufacturing of complex organic geometries

2024

Manish Aryal, Sourabh Deshpande, Jillian Aurisano, Sam Anand

ASME Manufacturing Science and Engineering Conference, <https://doi.org/10.1115/MSEC2024-121253>

Deep learning-based image segmentation for defect detection in additive manufacturing: an overview

2024

Sourabh Deshpande, Vysakh Venugopal, Manish Kumar, Sam Anand

Int. Journal of Advanced Manufacturing Technology, <https://doi.org/10.1007/s00170-024-14191-6>

Smart monitoring and automated real-time visual inspection of a sealant applications

2023

Sourabh Deshpande, Aditi Roy, Joshua Johnson, Ethan Fitz, Manish Kumar, Sam Anand

SME Manufacturing Letters, <https://doi.org/10.1016/j.mfglet.2023.08.115>

IIoT based framework for data communication and prediction using augmented reality for legacy machine artifacts

2023

Sourabh Deshpande, Shailesh Padalkar, Sam Anand

SME Manufacturing Letters, <https://doi.org/10.1016/j.mfglet.2023.08.058>

IIoT based augmented reality for factory data collection and visualization

2021

Jonathan Rosales, Sourabh Deshpande, Sam Anand

Procedia Manufacturing, <https://doi.org/10.1016/j.promfg.2021.06.062>

Prediction of selective laser melting part quality using hybrid Bayesian network

2020

Nathan Hertlein, Sourabh Deshpande, Vysakh Venugopal, Manish Kumar, Sam Anand

Additive Manufacturing, <https://doi.org/10.1016/j.addma.2020.101089>

Developing an open-source, inexpensive, large-scale polar configuration 3D printer 2019
Sourabh Deshpande, Shrinivas Kulkarni, Sid Shah, John Irwin
Int. Journal of Eng. Res. and Innov., <https://ijeri.org/IJERI-Archives/issues/spring2019/spring2019.htm>

Hybrid polymer AM of a Darrieus type vertical axis wind turbine design to improve power generation efficiency 2016
Sourabh Deshpande, Nithin Rao, Nitin Pradhan, John Irwin
ASME International Mech. Engineering Congress and Exp., <https://doi.org/10.1115/IMECE2016-65910>

RESEARCH AND INDUSTRY GRANT PROPOSALS

Assisted advisor Prof. Sam Anand in the preparation of the following down-selected research grants:

Human Digital Twin for Improving Worker Safety in Manufacturing and Logistics 2024
Bureau of Workers' Compensation, PI: Prof. Manish Kumar, Co-PI: Prof. Sam Anand
Ohio State Funding: **\$1.3M, UC Share of \$850K**

Human Digital Twin from Estimation to Human Motion Prediction, Ergonomics, and LLM 2024
University of Cincinnati Industry 4.0/5.0 Institute, PI: Prof. Sam Anand
Industry-Academia Consortium Project: **\$80K**

IIOT for Legacy and Smart Factory Machines with LLM feedback and Digital Twin 2024
University of Cincinnati Industry 4.0/5.0 Institute, PI: Prof. Sam Anand, Co-PI: Prof. Ming Tang
Industry-Academia Consortium Project: **\$40K**

Post-Secondary Workforce Development in the Healthcare and Manufacturing Industries 2022, 2023
RAPIDS Proposal for Southwest Ohio, PI: Prof. Sam Anand
Ohio Department of Higher Education State Funding **\$60K** (2022), **\$35K** (2023)

SMART VISTA – Visual Inspections Perfected using AI 2021
MxD, US Department of Army, Project PI: Dr. Aditi Roy, Siemens Technology
UC PI: Prof. Sam Anand, UC Co-PI: Prof. Manish Kumar
Federal Funding **\$1.5M**, UC Share of **\$150K**

RELEVANT COURSEWORK

- Intelligent Systems (AI)
- Statistical Methods
- Advanced Quality Engineering
- Math Methods for Decision Engineering
- Introduction to Industrial AI
- Computational Methods in Additive Manufacturing

ACADEMIC MENTORSHIP

Mentored M.S. students with their thesis completion

- Manish Aryal – ImVR: Enabling Immersive Exploration Environment for Complex Organic Geometries, 2023
- Anuj Gautam – Multi-modal LLM Framework for Smart Manufacturing, 2024 (Ongoing)

ACCOMPLISHMENTS AND LEADERSHIP EXPERIENCE

- University of Cincinnati Office of Research Digital Futures Fellowship, 2024
- Invention Disclosure (UC), along with PI Prof. Anand and Co-PI Prof. Kumar - Object Localization MxD, 2022
- Recognition by the SME and ASME for leading (student) in organizing NAMRC 49 and MSEC, 2021
- NSF student travel award for attending NAMRC/MSEC, 2021
- Graduate teaching assistant, MECH 6071, Advanced Design for Manufacturing, 2019
- University of Cincinnati graduate incentive scholarship, 2018 – 2024
- Gold and bronze medal at national level cycling tournament, School Games Federation of India, 2010
- Best All-Around and Best Character award, Tata Motors Employees Education Trust, India, 2008